Predicting Market Volatility and Short-Term Trading Signals from Reddit's WallStreetBets

UE24CS352A Machine Learning - Mini-Project write-up | Team: Prithvi Aithal (PES1UG24CS909), Sagar A Walikar (PES1UG24CS396) | Repository: github.com/Prithvi-Aithal/wsb-market-prediction (private)

1. Problem statement

Retail-investor communities such as r/WallStreetBets (WSB) moved prices during the January 2021 GameStop short squeeze. We ask whether the **posts themselves** contain a usable signal about (a) the **direction** of the next price move (will open(next day) exceed open(this day)?), (b) **volatility** (will the move be unusually large?), and (c) whether such predictions can drive a simple short-term trading strategy. We re-implement the CS229 report's approach, extend it with hyper-parameter tuning, stricter baselines, a leakage-free evaluation and a back-test, and add a **new model** (a drift-aware shrinkage ensemble, section 4) to improve on the report's two models.

2. Dataset

Posts: Kaggle *gpreda/reddit-wallstreetsbets-posts* - 53,187 posts (Sep 2020 - Aug 2021) with title, score, id, url, comment count, creation time, body, timestamp. **Prices:** daily OHLC from Yahoo Finance (*yfinance*). Posts mentioning each stock: GME 14,619, AMC 5,310, TSLA 1,232, FB 251 (ticker or company name). Each post becomes one example, as in the report.

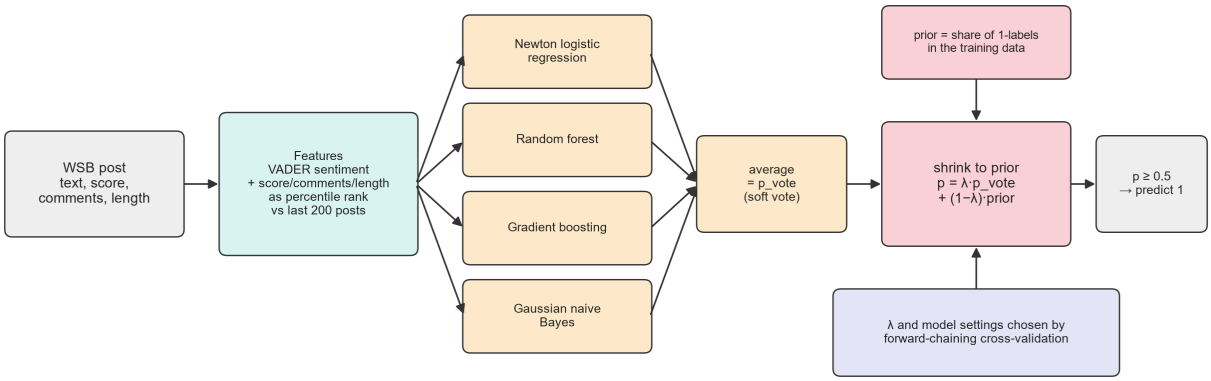
3. Approach (replicating the report)

Pre-processing: drop NaN titles, lowercase, remove URLs, handles, special and single characters, collapse spaces. **Features (7)**: post_score, total_comms_num, len_of_post, positive/neutral/negative sentiment (VADER), bias term; count features are log1p-transformed and standardised on the training set. **Labels**: *direction* = open(k+1) > open(k); *volatility* = |open(k+1)/open(k)-1| above the training-set median (the report's 'significant shift'). Two timing conventions: *paper* (k = post's trading day) and *tradable* (k = first open after the post, so no look-ahead). **Models**: (1) logistic regression solved with **Newton's method implemented from scratch** (small L2 term keeps the Hessian invertible); (2) 3x256 ReLU network, sigmoid output, Adam (lr 1e-3), 30 epochs, batch 16 (64 for GME/TSLA), best epoch by validation accuracy; (3) uniform-random and majority-class baselines. **Tuning**: logistic-regression L2 in {1e-6..10} by validation log-loss; network random search (6 configs: depth/width, learning rate, weight decay, batch) by validation accuracy, re-trained with 5 seeds. **Evaluation**: chronological 80/10/10 train/val/test split, so the test set is strictly in the future.

4. New model: drift-aware shrinkage ensemble (our addition)

Why the report's models fail. Training data is dominated by the squeeze month; the test months behave differently in two ways. *Label shift*: 'volatile' means a move above the training median, so 46% of GME training posts are volatile but only 19% of test posts. *Feature drift*: the subreddit grew, so GME's mean log post-score rises from 3.2 (train) to 5.8 (test), about 14x more upvotes. Models that learned the squeeze keep predicting it and fall far below the majority baseline. The new model has four parts:

(a) Relative features. Score, comments and length are replaced by their percentile rank among the stock's previous 200 posts (only earlier posts, so no look-ahead): 'top 10% of recent posts' means the same in February and in July. Sentiment is unchanged. **(b) Four models, soft vote.** Newton logistic regression, a random forest (bagging), gradient boosting (boosting) and Gaussian naive Bayes are trained on the same features; their probabilities are averaged with equal weights: p_vote. **(c) Shrinkage toward the prior.** The prior is the share of 1-labels in the training data, i.e. the best guess before reading the post (Bayes' prior). The final probability is $p = \lambda\, p_vote + (1-\lambda)$ prior: $\lambda = 0$ is the majority-class rule, $\lambda = 1$ trusts the ensemble fully, so λ sets the bias-variance trade-off. **(d) Tuning by forward-chaining cross-validation.** Random search over 8 configurations (L2; forest leaf size; boosting trees, depth, learning rate), each with λ in {0.0, 0.1, 0.25, 0.5, 0.75, 1.0}: 48 candidates. Each is scored by log-loss on 4 time-ordered folds: fit on the oldest 40/60/80% of training and score the next 20%, then fit on all training and score validation (a fold whose training part holds only one class is skipped, e.g. AMC direction: 3 folds). Lowest mean log-loss wins (ties to the smaller λ); the winner is refit on train+val and evaluated once on test. If the posts did not help predict the future in CV, λ goes to 0 and the model falls back to the prior.



5. Implementation overview

Python; *src/wsb_pipeline.py* (data, features incl. relative ranks, labels, splits), *src/models.py* (Newton logistic regression, network, *ShrinkageEnsemble* and *tune_ensemble*), *src/run_experiments.py* (experiments, tuning, baselines, back-test, figures), *src/leakage_check.py* (shuffled vs chronological split), *src/demo.py* (live scoring of a post, and *--quick*, which runs the whole pipeline live on one stock), *tests/* (12 unit tests). Back-test: each trading day, majority vote of the day's per-post predictions -> long

(open to next open) or flat, compared with going long on every day that had posts.

6. Results (held-out chronological test set, accuracy)

Direction task, 'tradable' labels

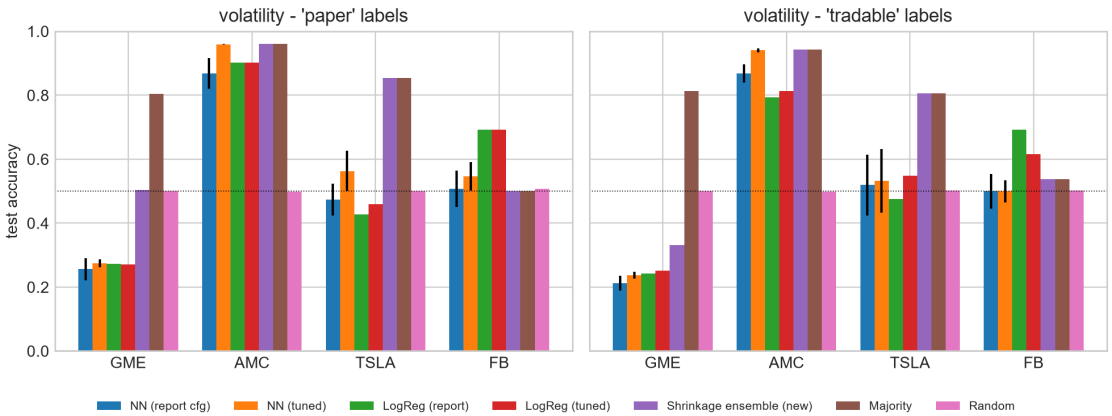
Stock	n test	NN (report)	NN (tuned)	LogReg (report)	LogReg (tuned)	Ensemble (new)	Majority	Random
GME	1462	0.569	0.569	0.567	0.567	0.568	0.567	0.500
AMC	531	0.350	0.353	0.411	0.388	0.439	0.653	0.500
TSLA	124	0.563	0.494	0.573	0.589	0.556	0.556	0.502
FB	26	0.500	0.469	0.462	0.538	0.577	0.577	0.495

Volatility task, 'paper' labels

Stock	n test	NN (report)	NN (tuned)	LogReg (report)	LogReg (tuned)	Ensemble (new)	Majority	Random
GME	1462	0.256	0.275	0.272	0.271	0.503	0.805	0.500
AMC	531	0.869	0.960	0.902	0.902	0.960	0.960	0.499
TSLA	124	0.474	0.563	0.427	0.460	0.855	0.855	0.500
FB	26	0.508	0.546	0.692	0.692	0.500	0.500	0.507

Leakage check (report's direction task, report configuration): with a *shuffled* split the network scores GME 0.77, AMC 0.78, TSLA 0.53, FB 0.68; on our *chronological* split it scores GME 0.57, AMC 0.35, TSLA 0.60, FB 0.47. Same-day posts share a label, so shuffling leaks the answer.

New model over all 16 settings (both label modes, both tasks): it matches or beats the majority baseline in **11/16** (best tuned old model: 6/16) and matches or beats both tuned old models in **12/16**. Mean accuracy: ensemble 0.605, tuned LogReg 0.542, tuned NN 0.520, majority 0.685. Mean test log-loss: ensemble 0.639, LogReg 0.667, NN 0.822, majority 0.662. CV chose $\lambda = 0$ (pure prior) in 4 settings; where $\lambda = 0$ the model equals the majority baseline. Still below it: AMC direction (paper), FB direction (paper), GME volatility (paper), AMC direction (tradable), GME volatility (tradable).



7. Conclusions

(i) **Honest evaluation matters.** We could not reproduce the report's high accuracies (GME 0.84, AMC 0.82) on a time-ordered split; a shuffled split inflates AMC and FB, which suggests leakage is a likely cause (we cannot verify the report's exact split). On a future hold-out the report's models do not beat the majority-class baseline on direction by more than the test-set standard error (AMC is far below it). (ii) **Tuning the report's models helps only modestly:** AMC volatility network 0.87 to 0.96 (only matching the majority baseline 0.96); TSLA 0.47 to 0.56 (still far below 0.85). (iii) **The new shrinkage ensemble is the most reliable model**: majority-level in 11/16 settings, with the largest gains where the old models failed worst (GME volatility 0.50 vs 0.28, TSLA volatility 0.85 vs 0.56, AMC direction 0.44 vs 0.39). It wins by *recognising when the posts do not generalise* and falling back to the prior; its test AUC stays near 0.5, so it finds no real predictive signal in the posts either. (iv) The back-tests show no edge: the network strategy stays flat on GME/AMC/TSLA, and neither strategy beats always-long. (v) Distribution shift (squeeze vs after) and tiny TSLA/FB test sets (124 and 26 posts, std. error 5-10 pp) limit what can be learned. **Future work:** richer text features (BERT embeddings), day-level aggregation, more data, and a transaction-cost-aware strategy.